INCH-POUND

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MILITARY SPECIFICATION

COUPLING ASSEMBLIES, QUICK DISCONNECT, AIRCRAFT LIQUID OXYGEN SYSTEMS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of liquid oxygen quick-disconnect coupling assemblies and coupling halves.

1.2 Classification. Coupling assemblies shall be of the following types, as specified (see 6.2b):

Type I - Supply coupling assemblies

Type II - Vent coupling assemblies

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein, 'Unless otherwise specifiect, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2d).

SPECIFICATIONS

FEDERAL

O-T-620	1, 1, 1, Trichloroethane, Technical, Inhibited (Methyl Chloroform)
PPP-B-566	Boxes, Folding, Paperboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 53), Naval Air Engineering Center, Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS (Cont'd)

FEDERAL

PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-665	Boxes, Paperboard, Metal Edged and Components
PPP-B-676	Boxes, Setup
PPP-C-1752	Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible
PPP-C-1797	Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam
PPP-C-1842	Cushioning Material, Plastic, Open Cell (For Packaging Applications)

MILITARY

MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-O-27210	Oxygen, Aviator's Breathing, Liquid and Gas

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-970	Standards and Specifications, Order of Precedence for The Selection of
MIL-STD-781	Reliability Testing For Engineering Development, Qualification, and Production
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-889	Dissimilar Metals
MS22068	Coupling Assemblies, Quick Disconnect, Aircraft Liquid Oxygen Systems

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094)

* 2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein.Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2d).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also my be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein, (except for related associated detail specifications, specification sheets, or MS standards) the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Qualification. The couplings furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.3, 4.3.1, and 6.4). In addition, the retention of the qualification for the couplings on the applicable qualified products list shall be dependent on periodic verification of continued compliance with the requirements of this specification (see 4.3 and 4.3.2).

* 3.2 First article. When specified, the coupling shall be subjected to first article inspection (see 6.5) in accordance with 4.4.

* 3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.4 Materials. Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.4.1 Metal parts. All metal parts shall be of a corrosion-resistant material or treated in a manner to render them adequately resistant to corrosion.

3.4.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.2 Protective treatment. When materials are used in the construction of the couplings that are subject to deterioration when exposed to environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of environmental conditions shall be avoided.

3.5 Design and construction. The design and construction of the coupling shall be in accordance with MS2206/3, The outer sleeve of the female coupling half shall rotate freely about the fitting end and center body of the female coupling half, in the uncoupled condition.

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* 3.5.1 Type I coupling. The type I coupling shall contain a check valve in each coupling half. each the coupling half When the coupling is connected, liquid oxygen shall flow. When the coupling is disconnected, the flow shall be checked automatically. The coupling shall contain a positive locking device that will automatically lock the connection of the coupling halves. A minimum of 1.0 pounds axial force shall be provided to maintain the locking device in the locked position. The axial load required to depress the positive locking device to fully disengage shall not exceed 9 pounds.

3.5.2 Type II coupling. The type II coupling shall contain no check valves. Gaseous oxygen shall flow through the coupling whether connected or disconnected.

* 3.5.3 Decreasing. All internal surfaces of the couplings shall be degreased in accordance with MIL-STD-1359. After completion of the cleaning and when . assembled, a halide testing apparatus having a sensitivity of at least 3 X 10^{-4} standard cc/see shall be used to determine the absence of the cleaning compound.

3.6 Performance.

3.6.1 Interchangeability.

3.6.1.1 Type I. When type i male and female coupling halves are connected respectively qualified female and male coupling halves and tested as specified in 4.7.2.1, the coupling assembly shall conform to the following:

(a) Torque, type I	(3.6.2.1)
(b) Locking device, type I	(3.6.3)
(c) Leakage, type I	(3.6.4)
(d) Pressure drop, type I	(3.6.5)

3.5.1.2 Type II. When type II male and female coupling halves are connected respectively to a qualified female and male coupling halves and tested as specified in 4.7.2.2, the maximum torque required to connect and disconnect the coupling halves shall not exceed 25 inch-pounds and the minimum disconnection torque shall be not less than 10 inch-pounds.

3.6.2 Torque.

3.6.2.1 Type I. The maximum torque required to connect and disconnect the coupling halves tested as specified in 4.7.3.1, shall not exceed 35 inch-pounds. The minimum torque required to disconnect the coupling halves shall be not less than 5 inch-pounds.

3.6.2.2 Type II. The maximum torque required to connect and to disconnect the coupling halves, when tested as specified in 4.7.3.2, shall not exceed 25 inch-pounds. The minimum torque required to disconnect shall be not less than 10 inch-pounds.

3.6.3 Locking device, type I.

* 3.6.3.1 Without torque applied. When tested as specified in 4.7.4.1, the axial force required to depress the positive locking device sufficiently to disengage it shall be a minimum of 1 pound to a maximum of 9 pounds.

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3.6.3.2 With torque applied. When testted as specified 4.7.4.2, the locking device shall not become disengaged.

3.6.4 Leakage, type I.

3.6.4.1 Coupling halves. The coupling leakage, when tested as specified in 4.7.5.1, shall not exceed 1 cubic centimeter (cc) per minute.

3.6.4.2 Coupling assembly. The coupling leafage, when tested as specified in 4.7.5.2, shall not exceed 1 cubic centimeter (cc) per minute.

3.6.4.3 Coupling assembly, with side load. The coupling leakage, when tested as specified In 4.7.5.3 shall not exceed cubic centimeter (cc) per minute.

3.6.5 Pressure drop, type I. The pressure drop, when tested as specified in 4.7.6, shall be in accordance with Table I.

3.6.6 Proof pressure, type I. The coupling assemblies, when tested as specified in 4.7.7, shall not show any evidence of distortion, failure or malfunction.

3.6.7 Operational test.

3.6.7.1 Type I. The coupling assembly leakage, when tested as specified in 4.7.8.1, shall not exceed 1.0 cc/min and the disconnecting and connecting torque shall be as specified in 3.6.2.1.

3.6.7.2 Type II. The disconnecting and connecting torque, when tested as specified in, shall be as specified in 3.6.2.2.

3.6.8 Burst pressure (static), type I. The coupling, when tested as specified in 4.7.9, shall not rupture.

3.6.9 High temperature.

3.6.9.1 Type I. The coupling assembly, when tested as specified in 4.7.10.1, shall not show any evidence of mechanical or material failure.

3.6.9.2 Type II. The coupling assembly, when tested as specified in 4.7.10.2, shall not show any evidence of mechanical or material failure.

3.6.10 Vibration.

3.6.10.1 Type I. The coupling assembly, when tested as specified in 4.7.11.1. shall not become disconnected during the test and shall not show any evidence-of mechanical or material failure.

3.6.10.2 Type II. The coupling assembly, when tested as specified in 4.7.11.2, shall not become disconnected during the test and shall not show any evidence of mechanical or material failure.

3.6.11 Salt fog.

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3.6.11.1 Type I. The coupling assembly, when tested as specified in 4.7.12.1, shall not show any evidence of corrosion or deterioration.

3.6.11.2 Type II. The coupling assembly, when tested as specified in 4.7.12.2, shall not show any evidence of corrosion or deterioration.

3.6.12 Reliability.

3.6.12.1 Type I. The coupling assembly, shall have a specified mean cycles between failure (MCB-F) of not less than 700 cycles at the 90 percent lower confidence limit. This is equivalent to a minimum acceptable MCBF of 466 cycles and to a reliability of .9986 for a one cycle mission.

3.6.12.2 Type II. The coupling assembly, shall have a specified mean cycles between failure (MCBF) of not less than 700 cycles at the 90 percent lower confidence limit. This is equivalent to a minimum cycle MCBF of 466 cycles and to a reliability of .9986 for a one cycle mission.

3.6.13 Disassembly and examination, types I and 11. When examined as specified in 4.7.14. the coupling assemblies shall not show evidence of excessive deterioration or wear as a result of testing which might affect the coupling performance.

* 3.7 Identification of product. Each coupling shall be permanently and legibly marked by stamping in letters and numbers 0.045 inch or larger in height with the manufacturer's name or code identification, Government part number and date of manufacture. The markings, except the date of manufacture, shall be stamped on the wrench flats or hex nuts on couplings without a sleeve lock. The date of manufacture can be stamped on any convenient surface. On couplings with a sleeve lock, the above markings shall be stamped on the sleeve upper surface and shall also include the phrase "DEPRESS RING TO UNLOCK."

3.8 Workmanship. The coupling assemblies shall be uniform in quality and shall be from irregularities or defects which could adversely affect performance, reliability, or durability. The coupling assemblies shall be free of oil, grease, fuels, water, dust, dirt, or any other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements {examinations and tests) as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 6. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the

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contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The examination and testing of the couplings shall be classified as follows:

a. Qualification inspection (see 4.3)
b. First article inspection (see 4.4)
c. Quality conformance inspection (see 4.5)

* 4.3 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in Table II that are applicable to the coupling type submitted for qualification inspection.

4.3.1 Qualification samples. Qualification samples shall consist of the following:

a. Four couplings of each type for which qualifications is desired.

b. A Statement of Certification (with supporting test data) from the manufacturer that each type of coupling submitted conforms to the reliability requirements cited in paragraphs 3.6.12, 4.7.13 and 6.4.1.

c. Two sets of manufacturer's drawings for each type coupling submitted (see 6.4.1).

Samples shall be forwarded to the test facility set forth in the letter of authorization to submit samples (see 6.4). The samples shall be plainly identified by securely attached durable tags marked with the following information:

> Samples submitted by (name) on (date) for qualification inspection in accordance with the requirements of MIL-C-21049F under authorization (reference authorizing letter by code, serial number, and date} (see 6.4).

4.3 .2 Retention. The retention of qualification shall consist of periodic verification to determined compliance of the qualified coupling with the requirements of this specification. The time and method of periodic verification shall be specified by the activity responsible for the Qualified Products List and shall be included in the Notice of Qualification letter.

* 4.4 First article inspection. First article inspection shall consist of the examinations and tests specified in Table III that are applicable to the coupling type submitted for first article inspection, listed sequence mandatory.

4.4.1 First article samples.Unless otherwisse specified, as soon as practicable, after the award of the contract or order, the manufacturer shall submit two coupling assemblies or two coupling halves or each type specified in the contract or order. The samples shall be representative of the construction,

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workmanship, components, and materials to be used during production. When a manufacturer is in continuous production of these units from contract to contract, submission of further first article samples on new contract may be waived at the discretion of the acquiring activity (see 6.2e). Approval of the first article samples or the waiving of first article inspection does not preclude the requirements of submitting to the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6.2f).

4.4.1.1 First article information. Upon completion of the first article inspection, the Government activity responsible for conducting this inspection program (see 6.2f) shall report to the contracting officer the results of the inspection program, with appropriate recommendation.

4.4.1.2 Disposition of samples. Upon completion of the first article inspection program, disposition of samples shall be as follows:

a. One approved sample shall be returned to the manufacturer for use in monitoring production.

b. One sample shall not be returned due to their having been destroyed during the test program. "

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in Table IV. The sampling and inspection levels and acceptance criteria shall conform to MIL-STD-105.

4.5.1 Sampling.

4.5.1.1 Inspection lot.

* 4.5.1.1.1 Coupling assembly. An inspection lot size shall be expressed in units of coupling assenmblies or coupling halves made essentially under the same conditions and from the same materials and components. The sample unit shall be one coupling assembly or coupling halve.

4.5.1.1.2 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing couplings fully prepared for delivery, made from essentially the same materials and components. The sample unit shall be one shipping container, containing couplings fully prepared for delivery with the exception that it need not be sealed.

4.5.1.2 Sampling for tests and examinations of the coupling assemblies. The sample size, acceptance criteria, tests, and examinations required for the coupling assemblies shall be as specified in Table IV.

4.6 Test conditions.

4.6.1 Oxygen. Oxygen used in testing the couplings shall be oxygen conforming to type I or type II, as applicable, of MIL-O-27210.

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4.6.2 Temperature and pressure. Unless otherwise specified, tests shall be conducted at local ambient temperature and barometric pressure. The temperature and barometric pressure shall be recorded at the time of inspection. This information shall be available for computation of test data, where required, to normal temperature and pressure (NTP) conditions. NTP conditions are 29.92 inches of mercury and 70 F. Test instruments shall be calibrated or adjusted according to their required usage in conducting individual tests.

4.7 Inspection methods.

4.7.1 Visual examination.

4.7.1.1 Coupling assembly. Every coupling assembly shall be examined visually for defects to determine conformance to this specification. The classification of defects, Table V, shall be used to classify the defects found.

4.7.1.2 Dimensions. The couplings shall be checked dimensionally to determine conformance to the dimensions specified in MS22068 and the automatic locking of the locking device when the type I coupling halves are connected. when measuring the travel of the lock sleeve, a constant angle shall be maintained between the diametrical plane of the lock sleeve and the longitudinal centerline of the coupling.

4.7.1.3 Packaging. Each of the fully prepared shipping containers, containing couplings selected as a sample unit from the lot shall be examined to determined that the packaging, packing and marking conform to this specification. The classification of defects, Table VI shall be used to classify the defects found.

4.7.2 Interchangeability.

4.7.2.1 Type I. The type I female coupling half shall be connected to each type I male coupling half that has been approved for listing on the Qualified Products List (QPL). Each coupling assembly shall then be subjected to the following tests:

Torque, type I (4.7.3.1)	Leakage, type I (4.7.5)
Locking device, type I (4.7.4)	Pressure drop, type I (4.7.6)

The above procedure shall then be repeated using the type I male coupling half with the female coupling halves that have been approved for listing on the QPL. Each combination coupling assembly shall pass the requirements specified in 3.6.1.1.

4.7.2.2 Type II. The type II female coupling half shall be connected to each type II male coupling half that has been approved for listing on the Qualified Products List (QPL). Each coupling assembly shall then be subjected to the torque type II test (4.7.3.2). The above procedure shall then be repeated using the type II male coupling half with the female coupling halves that have been approved for listing on the QPL. Each combination coupling assembly shall pass the requirements specified in 3.6.1.2.

4.7.3 Torque.

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4.7.3.1 Type 1. With the male and female halves connected and the outlet port capped, a-pressure of 120 psig gaseous oxygen shall be applied to the inlet port. Mith this pressure maintained, and the locking device disengaged, the torque required to disconnect and connect the coupling halves shall be measured for five (5) cycles. One cycle shall consist of a disconnection and connection of the coupling halves. The torque required to disconnect and connect the coupling halves shall pass the requirements specified in 3.6.2.1.

4.7.3.2 Type II: The torque required to disconnect and connect the coupling halves shall be measured for five (5) cycles. One cycle shall consist of a disconnection and connection of the coupling halves.The torque required to disconnect and connect the coupling halves shall pass the requirements specified in 3.6.2.2.

4.7.4 Locking device, type I.

* 4.7.4.1 Without torque applied. With the coupling halves connected and the locking device engaged, the axial force required to depress the positive locking device sufficiently to disengage it shall conform to the requirements specified in 3.6.3.1.

4.7.4.2 With torque applied. With the coupling halves connected and the locking device engaged, a torque of 35 inch pounds shall be applied to the coupling assembly in the direction of disconnection. The locking device shall pass the requirements specified in 3.6.3.2.

4.7.5 Leakage, type I.

4.7.5.1 Coupling halves. The inlet end of the male and the outlet end of the halves shall each be connected to a gaseous oxygen source capable of providing a pressure up to 120 psig. With the coupling halves disconnected, pressures of 75 and 120 psig shall be applied to each coupling half. Each pressure shall be applied for a minimum of 3 minutes and leakage measured. The coupling shall pass the requirements specified in 3.6.4.l.

4.7.5.2 Coupling assembly. With the coupling halves connected, a pressure of 75 and 120 psig shall be applied to the inlet end of the male coupling with the outlet of the female coupling capped. Each pressure shall be applied for 3 minutes and leakage measured. The coupling shall pass the requirements specified in 3.6.4.2.

4.7.5.3 Coupling assembly, with side load. The coupling assembly shall be mounted to a test fixture In the manner shown in figure 1. The coupling assembly shall be in a vertical position with the male fitting end facing down and firmly connected to the test fixture. A pressure of 100 + 10 psig shall be applied through the male fitting end. While maintaining this pressure, a 5 pound side load shall be applied to the upper edge of the capped outlet of the female coupling in the location and directions shown in figure 1. The load shall be maintained for a minimum of 1 minute in each direction and the leakage measured. The coupling shall pass the requirements specified in 3.6.4.3.

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4.7.6 Pressure drop, type I. With the coupling halves connected, the coupling inlet shall be connected to a pressure source of 20 psig gaseous oxygen. With an inlet pressure of 20 psig, the pressure drop through the coupling assembly shall be determined for flow rates of 10, 60 and 120 liters per minute. The coupling assembly shall pass the requirements specified in 3.6.5.

4,7.7 Proof pressure, type I. With the coupling halves connected and the outlet port sealed, a pressure of 200 psig gaseous oxygen shall be applied to the inlet port of the coupling assembly. The coupling assembly shall pass the requirements specified in 3.6.6.

4.7.8 Operational test.

4.7.8.1 Type I. A liquid oxygen source of 60 to 90 psig shall be supplied to the inlet of the coupled assembly. The outlet shall have 6 to 12 inches of tubing connected to a manual valve. After starting a flow of liquid oxygen, submerge the cold coupling assembly in a bath of O-T-620 (trichloroethane) and flow for two minutes. After assuring that trapped air is released from the coupling, measure coupling leakage by the bubble collection method. Remove the assembly from the bath and immediately measure the torque required to disconnect and connect the coupling halves. Repeat the above procedure two additional times. The coupling assembly shall pass the requirements specified in 3.6.7.1. At the completion of this test, the coupling assembly shall be subjected to and pass the leakage, test specified in 4.7.5.

4.7.8.2 Type II. The type II coupling assembly shall be connected to a liquid oxygen source and a flow of liquid oxygen produced through the coupling. The flow shall be stopped and the coupling halves immediately disconnected. The process of connecting the halves, producing a flow of liquid through the coupling, stopping the flow, and disconnecting the halves shall be repeated five times and the torque measured for each connection and disconnection. The coupling assembly shall pass the requirements specified in 3.6.7.2.

4.7.9 Burst pressure (static), type I. With the outlet port of the coupling sealed, a gaseous oxygen pressure of 350 psig shall be applied to the inlet port of the- couplinq and maintained for 15 seconds. The coupling shall pass the requirements specified in 3.6.8.

4.7.10 High temperature.

4.7.10.1 Type I. The coupling assembly shall be placed in a chamber having an internal temperature of 260 + 5°F (127 + 3°C) for a minimum period of 1 hour. After this exposure, the coupling shall be-removed from the chamber and immediately subjected to and pass the torque, type I (4.7.3.1) and leakage, type I (4.7.5) tests. The coupling assembly shall pass the requirements specified in 3.6.9.1.

4.7.10.2 Type II. The high temperature test shall be the same as for type I except that the coupling assembly shall be subjected only to the torque, type II test (4.7.3.2) after temperature conditioning. The coupling assembly shall pass the requirements specified in 3.6.9.2.

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4.7.11 Vibration. The coupling assembly shall be subjected to a random vibration test in accordance with MIL-STD-810, Method 514.3, Procedure I, Category 5, except the test conditions shall be in accordance with 4.6.2. W_o shall be 0.30 G^2/Hz and the test duration shall be one hour in each axis.

4.7.11.1 Type I. The coupling assembly shall be tested, as specified in 4.7.11, with the coupling assembly outlet capped and a pressure of 120 to 150 psig applied to the coupling assembly inlet for approximately one-half the test time in each axis. The remaining test time shall be performed with the coupling assembly outlet open and a pressure of 60 to 90 psig applied to the coupling assembly inlet with a cryogenic gas of a temperature to establish an internal coupling temperature or -250 F or colder. During this vibration, a torque of 35 inch-pounds shall be intermittently applied in a disconnect direction at intervals of five minutes. The coupling assembly shall pass the requirements specified in 3.6.10.1. Upon completion of vibration, the coupling assembly shall be subjected to, and pass the torque, type I (4.7.3.1) and the leakage, type I (4.7.5) tests.

4.7.11.2 Type II. The coupling assembly shall be tested, as specified in 4.7.11, with no internal pressure for approximately one-half the test time in each axis. The remaining test time shall be with a flow of cryogenic as of a temperature to establish an internal coupling temperature of -250 F or colder. The coupling assembly shall pass the requirements specified in 3.6.10.2. Upon completion of the vibration test, the coupling assembly shall be subjected to, and pass the torque, type II test (4.7.3.2).

4.7.12 Salt fog..

4.7.12.1 Type I. The coupling assembly shall be subjected to a salt fog test in accordance MIL-STD-810, Method 509.2, Procedure I, except the salt fog exposure period shall be 168 hours. The coupling assembly shall pass the requirements specified in 3.6.11.1. After completing the salt fog test, the coupling assembly shall be subjected to and pass the following tests:

Torque, type I (4.7.3.1)
Locking device, type I (4.7.4)
Leakage, type I (4.7.5)

4.7.12.2 Type II. The salt fog test shall be the same as for type I, except the coupling assembly shall be subjected only to the torque, type II test (4.7.3.2) after the salt fog test. The coupling assembly shall pass the requirements specified in 3.6.11.2.

4.7.13 Reliability.

4.7.13.1 Type I. The coupling assembly shall be subjected to cycles of disconnections and connections. A cycle shall consist of a disconnection (coupling halves separated) and a connection with the coupling outlet port sealed and 120 psig gaseous oxygen pressure applied to the coupling inlet port. The coupling assembly shall be subjected to the torque, type I (4.7.3.1), locking device, type I (4.7.4) and the leakage, type I (4.7.5) tests every 700 cycles during the reliability test to determine if a failure has occurred. A coupling assembly failing any of these tests shall have failed the reliability test. In

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addition, any failure of the coupling to mate properly or fracture of components shall constitute a test failure. No parts shall be replaced during the reliability testing as preventive maintenance. Acceptance shall be based on the accept-reject criteria of test plan I of MIL-STD-781. Recording, data handling, and reporting procedures shall be in accordance with MIL-STD-781.

4.7.13.2 Type II. Reliability testing shall be the same as for type I except that the coupling assembly shall be unpressurized during the test and shall be subjected only to the torque, type II test (4.7.3.2) every 700 cycles.

4.7.14 Disassembly and examination types I and II. At the conclusion of all tests specified herein or after any test in which the coupling failure indicates faulty construction, the coupling assembly shall be disassembled and examined. The coupling assembly shall pass the requirements specified in 3.6.13.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or C, as specified by the acquiring activity (see 6.2g).

* 5.1.1 Levels A&B. Each coupling shall be preserved and packaged in accordance with method ICI of MIL-P-116 without the use of contact preservative. Cleaning shall be accomplished in accordance with MIL-STD-1359. Cleaning by immersion shall not be permitted on the completed coupling assembly. The bagged coupling shall be cushioned with material conforming to PPP-C-1842., PPP-C-1797, or PPP-C-1752, minimum thickness of one inch and then enclosed within a heat sealed barrier bag fabricated from material conforming to MIL-B-121, Grade A. Prior to sealing all excess air must be exhausted. Each coupling shall then be packaged within an individual close-fitting box conforming to PPP-B-566, PPP-B-665 or PPP-B-676. The closure shall be accomplished in accordance with the procedures specified in the appendix of the applicable specification.

5.1.2 Level C. Couplings shall be packaged in a manner that will prevent physical damage or environmental deterioration during transit from the shipping point to the original consignee.

5.2 Packing. Packing shall be Level A, B or C as specified by the acquiring activity (see 6.2g). Insofar as practicable shipping containers shall effect a close fit, contain identical quantities and be of uniform dimensional configuration.

* 5.2.1 Level A. Couplings packaged as specified in 5.1.1 shall be packed for shipment in fiberboard shipping containers conforming to PPP-B-636, class-weather resistant Grade V3s/V3c. The gross weight of each packed container shall not exceed 70 pounds. The closure, sealing and reinforcing (banding) of the containers shall be accomplished in accordance with procedure specified in the appendix of the container specification.

5.2.2 Level B. Couplings packaged as specified in 5.1.1 shall be packed for shipment as specified in 5.2.1 except that the fiberboard shipping containers conforming to PPP-B-636, shall be Class-Domestic.

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5.2.3 Level C. Couplings packaged as specified, shall be packed for shipment in a manner that will afford protection against damage during transit from the shipping point to the original consignee. Containers shall be in accordance with Uniform Freight Classification Rules or regulations or other carriers applicable to the mode of transportation utilized.

* 5.3 Marking. Each unit and shipping container shall be marked in accordance with MIL-STD-129. In addition, the following precautionary marking shall be required on each unit package and shipping container (see 6.2i):

CAUTI0N

DO NOT STORE IN PROXIMITY OF PETROLEUM PRODUCTS

DO NOT ALLOW CONTAMINANTS OF ANY KIND TO BE USED ON OR ABOUT THE OXYGEN COUPLINGS

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The couplings are intended for use in aircraft liquid oxygen systems to uncouple the line without draining the system and without undue loss of liquid oxygen.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification (including amendments).

b. Types of coupling assembly or coupling half required (see 1.2).

c. Government part number and quantity.

d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

e. Whether first article inspection if waived (see 4.4.1).

f. Name and address of the first article inspection test facility (see 4.4.1) and the name and address of the Government activity responsible for conducting the first article inspection program (see 4.4.1.1 and 6.5).

g. Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

h. Applicable methods of cleaning and preservation.

i. Whether any special markings are required (see 5.3).

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* 6.3 Consideration of data requirements. The following data reequirements should be considered when this specification is applied on-a contract. The applicable Data Item Description (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirement List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.4.1.1	DI-T-5329	First article inspection report	Use contractor format
4.7.13	DI-T-5329	Reliability Inspection Reports	Use contractor format

The above DID's were those cleared as of the date of this specification. The current use of DOD 5010.12-L Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL-21049) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they should be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is Commander, Naval Air Systems Command, Washington, DC 20361; however, information pertaining to qualification of products should be obtained from the Commander, Naval Air Development Center, Warminster, Pennsylvania 18974-5000, Attention: Code 6031.

6.4.1 Documentation. When requested, the manufacturer shall make available engineering drawings and inspection reports in accordance with SD-6.

6.5 First article. When a first article inspection is required, the item(s) should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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* 6.6 Subject term (key word) listing.

Breathing gas
Liquid oxygen systems
Locking device
Oxygen
Quick disconnect
supply coupling
Vent coupling

6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indlcate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations.Bidders and contractors are cautioned to evaluate the requirement of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Navy,- AS
Air Force - 11

Preparing activity:
Navy - AS
[Project No. 1660 - 0579)

Review Activities:
Air Force - 99, 71

TABLE I. Pressure drop.

Maximum pressure drop (inches - Hg)	Flow (liters per minute)
0.20	10
3.5	60
15.0	120

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TABLE II. Qualification inspection.

Inspection	Requirement Paragraph(s)	Test Method Paragraph(s)
Visual examination, types I and II	3.6.1.1	4.7.1.1
Dimensional examination, types I and II	3.6.1.2	4.7.1.2
Interchangeability, type I	3.6.1.1	4.7.2.1
Interchangeability, type II	3.6.1.2	4.7.2.2
Torque, type I	3.6.2.1	4.7.3.1
Torque, type II	3.6.2.2	4.7.3.2
Locking device, type I (without torque)	3.6.3.1	4.7.4.1
Locking device, type I (with torque)	3.6.3.2	4.7.4.2
Leakage, type I (coupling halves)	3.6.4.1	4.7.5.1
Leakage, type I (coupling assembly)	3.6.4.2	4.7.5.2
Leakage, type I (with side load)	3.6.4.3	4.7.5.3
Pressure drop, type I	3.6.5	4.7.6
Proof pressure, type I	3.6.6	4.7.7
Operational test, type I	3.6.7.1	4.7.8.1
Operational test, type II	3.6.7.2	4.7.8.2
Burst pressure, type I	3.6.8	4.7.9
High temperature, type I	3.6.9.1	4.7.10.1
High temperature, type II	3.6.9.2	4.7.10.2
Vibration, type I	3.6.10.1	4.7.11.1
Vibration, type II	3.6.10.2	4.7.11.2
Salt fog, type I	3.6.11.1	4.7.12.1
Salt fog, type II	3.6.11.2	4.7.12.2
Reliability, type I	3.6.12.1	4.7.13.1
Reliability, type II	3.6.12.2	4.7.13.2
Disassembly and examinations, types I and II	3.6.13	4.7.14

TABLE III. First article inspection.

Inspection	Requirement Paragraph(s)	Test Method Paragraph(s)
Interchangeability, type I	3.6.1.1	4.7.2.1
Interchangeability, type II	3.6.1.2	4.7.2.2
Visual examination, types I and II	3.6.1.1	4.7.1.1
Dimensional examination, types I and II	3.6.1.2	4.7.1.2
Torque, type I	3.6.2.1	4.7.3.1
Torque, type II	3.6.2.2	4.7.3.2
Locking device, type I (without torque)	3.6.3.1	4.7.4.1
Locking device, type I (with torque)	3.6.3.2	4.7.4.2
Leakage, type I (coupling halves)	3.6.4.1	4.7.5.1
Leakage, type I (coupling assembly)	3.6.4.2	4.7.5.2
Leakage, type I (with side load)	3.6.4.3	4.7.5.3
Proof pressure, type I	3.6.6	4.7.6
Operational, type I	3.6.7.1	4.7.8.1
Operational, type II	3.6.7.2	4.7.8.2
Burst pressure, type I	3.6.8	4.7.9

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TABLE IV. Quality conformance inspection.

INSPECTION	METHOD	SAMPLE SIZE	ACCEPTANCE CRITERIA
Visual examination, type I or II (See classification of defects)	4.7.1.1	a. Every coupling critical defects. b. Inspection Level II for minor defects 1/	a. Reject all units with any critical defects. b. An acceptable quality level of 2.5 defects per hundred units for minor defects
Dimensions, type I or II	4.7.1.2	Inspection level S-2 1/	Acceptance number zero, rejection number 1
Torque, type I or II	4.7.3.1 or 4.7.3.2	Every coupling	Reject all defective units
Leakage, Type I, coupling halves	4.7.5.1	Every coupling	Reject all defective units
Leakage, Type I, coupling assy.	4.7.5.2	Every coupling	Reject all defective units
Leakage, Type I, coupling assy. with side load	4.7.5.3	Inspection level S-2 1/	Acceptance number zero, rejection number 1
Pressure drop type I	4.7.6	Inspection level S-2 1/	Acceptance number zero, rejection number 1
Proof pressure type I	4.7.7	Inspection level S-2 1/	Acceptance number zero, rejection number 1
Operational test, type I or II	4.7.8.1 or 4.7.8.2	Inspection level S-2 1/	Acceptance number zero, rejection number 1
Packaging	4.7.1.3	Inspection level S-2 1/	Total acceptable quality level of 4.0 percent defective

1/ The sample size shall be based only on the applicable sample size code letter corresponding to the specified inspection level of MIL-STD-105.

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TABLE V. Classification of defects for visual examination of the coupling assemblies.

Critical	Minor
1. Material imperfection or not as specified.	201. Marking missing, insufficient, incorrect, illegible, or not permanent.
2. Surface unclean, rough, misaligned, containing cracks, nicks, or other flaws.	
3. Any component missing, malformed, fractured, or otherwise damaged.	
4. Any component loose or not securely retained.	
5. Incorrect assembling or improper positioning of components.	
6. Any functioning part that works with difficulty.	
7. Locking device of type I does not go in lock detents when coupling halves are connected.	
8. Internal passage through the type II assembly is obstructed.	
9. Dimensions of threads not as specified.	
10. Faulty workmanship.	

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TABLE VI. List of defects for packaging.

Item	Defects
Exterior and interior markings	Missing, incorrect, incomplete, illegible; or improper size, location, sequence, or method or application; markings not the same on the interior containers.
Packaging and packing materials	Any nonconforming component; any component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of the unit package, intermediate package, container flaps, loose strappings, etc.; bulging or distortion of the containers.
Exterior and interior weight or content	Number per container is more or less than required; gross or net weight exceeds the requirements; more than one type in the same container.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-21049F	31 DECEMBER 1990

3. DOCUMENT TITLE

Coupling Assemblies, Quick Disconnect, Aircraft Liquid Oxygen Systems

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Systems Engineering and Standardization Department	(201) 323-1280	624-1280
c. ADDRESS (Include Zip Code) Naval Air Engineering Center Lakehurst, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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